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R.A.V.I.K.A.N.T: Recognizing Ancient Visual Indus Knowledge using Advanced Neural Techniques

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Abstract. The Indus Valley Civilization is still among the most mysterious ancient societies, mostly because of its undeciphered writing. Conventional techniques of archaeology have failed to find the cognitive and linguistic patterns under these inscriptions. In this work, we present R.A.V.I.K.A.N.T. (Recognizing Ancient Visual Indus Knowledge using Advanced Neural Techniques), a deep learning-based method to examine and categorize Indus script symbols. We have presented a methodology for efficient image clustering and classification using deep learning techniques. The process begins with dataset loading and preprocessing, followed by feature extraction using ResNet50 and dimensionality reduction via Principal Component Analysis (PCA). We apply and evaluate clustering algorithms such as K-Means, Agglomerative, Spectral, and BIRCH using performance metrics like the Silhouette Score and Davies-Bouldin Index. The best-performing clustering method is selected, and cluster labels are assigned, ensuring sufficient cluster sizes through augmentation. Lastly, we train and test several CNN architectures (ResNet50, DenseNet121, EfficientNet, MobileNetV2, VGG16) using accuracy and loss metrics to find the best one for classification. Among the clustering algorithms evaluated, K-Means demonstrated the best performance with a Silhouette Score of 0.1331 and a Davies-Bouldin Index of 1.7191, making it the optimal choice for cluster assignment. We found that VGG16 did better than the other four models in validation accuracy (0.8475) and loss (0.4567) using K-5 fold, showing strong results in classification.

Keywords: Convolutional Neural Network (CNN), dimensionality reduction, K-Means, Agglomerative, Spectral, BIRCH, Silhouette Score and Davies-Bouldin Index

1. Introduction

One of the first urban civilizations, the Indus Valley Civilization (IVC), left an undeciphered collection of symbols and inscriptions. These visual depictions, often

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found on ceramics and seals, are believed to carry substantial cultural, linguistic, and cognitive knowledge. Although they have not produced clear answers, conventional methods of decoding the Indus script have mostly depended on linguistic and statistical investigations [1, 2]. New insights on ancient symbolic knowledge come from recent developments in artificial intelligence (AI) and neuroscience. Neuroscience studies of the mind and brain can help explain how the brain understands these signals, especially those focusing on visual cognition, attention mechanisms, and memory encoding. Indus writing also employs attention-based Artificial intelligence models, such as Transformers, are utilized to identify patterns and contextual links. These models have changed natural language processing (NLP). The purpose of this study is to look into how attention-based AI models combined with a neuroscientific approach could help us understand how people read and understand Indus script. In our paper, we labeled the dataset with clusters and then classified it with attention-based deep learning models.

1.1 Visual Cognition and Symbol Recognition

The human brain is designed to identify patterns and handle symbolic data. While object and letter recognition depend on the inferior temporal cortex (IT) and fusiform gyrus, shape decoding is mostly dependent on the visual cortex (V1–V5). Like deep learning networks that analyze pictures in layers, studies on hierarchical visual processing show that the brain gradually removes meaning from basic visual inputs.

In ancient scripts, the Indus symbols—which have complex patterns and recurring motifs—probably triggered top-down and bottom-up attention systems. Bottom-up processes, which are driven by stimuli, help us recognize basic forms. Top-down processes, which are controlled by thought, let us understand things at a higher level and are the building blocks of cognitive-perceptual learning. We have shown the connection of visual cognition and symbol recognition in Figure 1 [3].

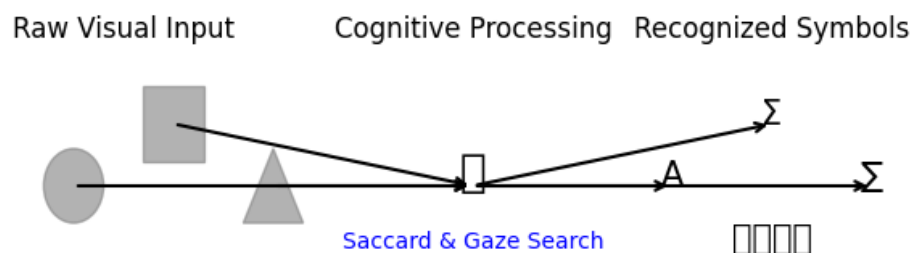


Figure 1: Cognitive and Recognized Symbol Relation

1.2 Neuroscientific Support of AI-Based Symbol Recognition

Researchers use neuroimaging techniques to make sure that AI-driven pattern recognition works with human thought processes. They focus on measuring cognitive load using EEG to better understand how challenging it is for a person to figure out

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old symbols. While the fMRI research on visual attention is used to track brain activity while you decode strange writings [4]. There are monitors for tracking the saccade movements in eye-tracking research to help readers better comprehend symbol scanning. By looking at the differences between what AI produces and how humans think and feel, we can make AI models that better reflect how humans think about symbols [5-7].

2 Related Work

When we discuss the Indus Valley script, we use a different set of deep learning models to sort the data into groups based on different feature extraction mechanisms. The work also uses pre-stopping criteria and focuses on advanced ideas like augmentation to be more efficient and faster than before. Additionally, the work employs pre-stopping criteria and emphasizes cutting-edge concepts like augmentation to be quicker and more effective than previously.

Several studies have explored the use of deep learning and computer vision for ancient script recognition and archaeological analysis. Many people have used **Convolutional Neural Networks (CNNs) and transformer-based models** to figure out patterns in ancient writings and symbols [8]. Research on deciphering the Indus script has primarily relied on statistical analysis and traditional machine learning with limited integration of advanced neural techniques. Recent works leveraging feature extraction methods like ResNet and clustering techniques have improved the classification and organization of ancient visual data. The various studies use different parameters, as illustrated in Table 1.

Table 1: Comparative Analysis of Papers Focusing on the Indus Scripts (✓ = Yes, ✗ = No) [9-13]

Parameter	Information 2023 [9]	Identifying Anomalous Indus Texts [10]	Handwritten Indus Signs Dataset [11]	Indus Script as an Alphabet [12]	Indus Script as an Alphabet [13]
Handwritten Text Recognition	✗	✗	✓	✗	✗
Deep Learning Models Used	✓	✓	✓	✗	✗
CNN for Text Recognition	✓	✗	✓	✗	✗

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Traditional Feature-Ba se OCR	✓	✓	✗	✓	✓
NLP Integration	✓	✓	✗	✓	✓
Data Augmentat ion for OCR	✗	✗	✓	✗	✗
Comparati ve OCR Study	✗	✗	✓	✗	✗
Pre-trained Model Usage	✓	✓	✓	✗	✗

3 Proposed Methodology

The suggested approach begins by loading the data, followed by picture preparation techniques such as scaling and normalizing. After ResNet50 is used to extract features, PCA is used to reduce the number of dimensions while keeping the 50 most important components. Similar photos are grouped through various clustering techniques, including K-Means, Agglomerative, Spectral, and BIRCH. The quality of the clustering is evaluated using the Silhouette Score and Davies-Bouldin Index to determine the best clustering method. Next, we convert the created clusters into labels and use different CNNs for further processing.

We use Albumentations for data augmentation on labels with fewer than 200 images, bringing the count up to 200 per label. Figure 2 shows the detailed steps. We have used two different evaluation metrics, loss and accuracy, for the comparison of the models performance. We assessed VGG16, ResNet50, DenseNet121, EfficientNet, and MobileNetV2 based on their structural differences and representative performance results. VGG16 is the simplest; ResNet50 computes deep learning through residual connections; DenseNet121's interconnectivity encourages feature reuse; EfficientNet promotes performance without compromising efficiency; and MobileNetV2 is trained with small footprint specifications. This incorporates a hierarchy of depth, ways to connect the structures, and performance through computational cost to rigorously inform the research question.

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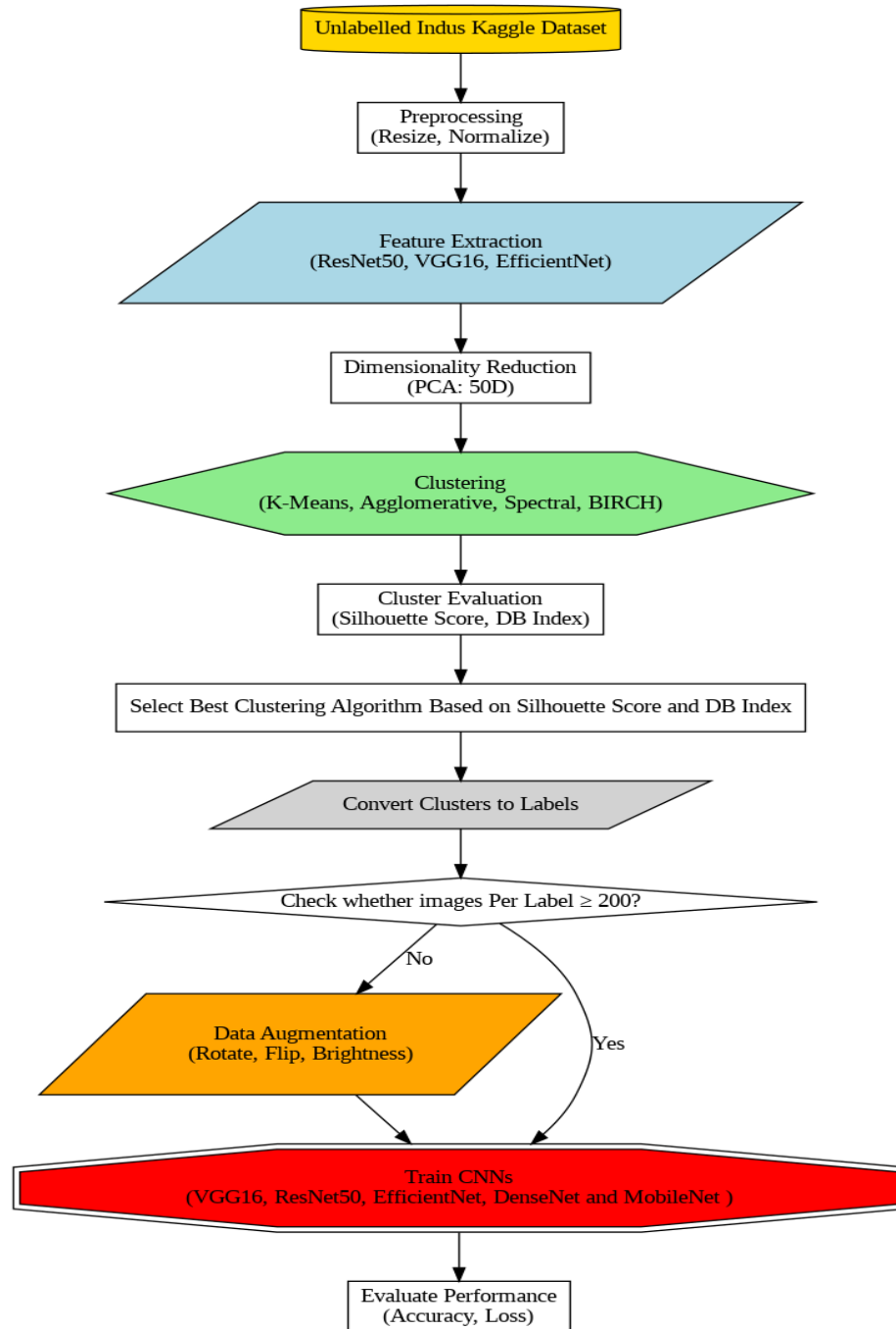


Figure 2: Proposed Methodology

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4 Materials and Methods

Python was used in this research to implement and test different clustering algorithms. The experiments were performed on a GPU T4 x2 platform, and the total running time took 20m 49s. The training and testing of the models were done using appropriate libraries like Scikit-learn and TensorFlow. Computational resources allowed for effective processing and analysis of the clustering performance metrics.

4.1 Dataset Used

We have used a Kaggle repository in our work. The Kaggle repository uses the collection of images and labels designated as "IM-417-150," which comprises the 417 original characters cataloged in "The Indus Scripts: Texts, Concordance, and Tables," published by Iravatham Mahadevan. In the dataset, there are two parts: 1. Grayscale Images 2. The pixel intensity values of the grayscale images were analyzed. The images did not have any labeled categories, so we used machine learning algorithms to cluster them accordingly. We used KMeans, Agglomerative, ADBSCAN, Spectral, and BIRCH clustering in our work. Sample images of the dataset are shown in Figure 3.

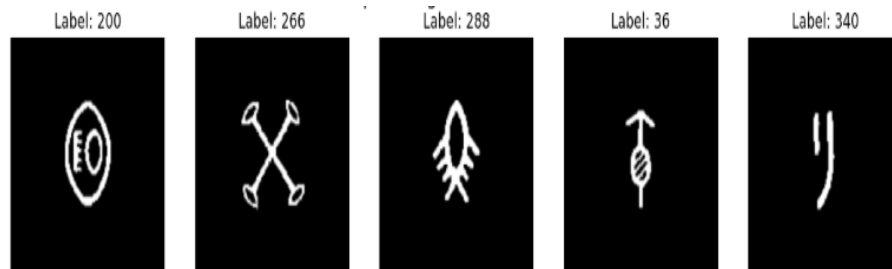


Figure 3: Sample grayscale pictures taken from the Indus script collection used in this work. Every image stands for a different symbol taken from Iravatham Mahadevan's public dataset. These images are used as the input for jobs in classification and clustering.

We also used different evaluation metrics to find out the most suitable machine learning algorithm for our specific task. Among all the algorithms, K-means performed better with the scores. Silhouette Score: 0.13310441374778748, Davies-Bouldin Index: 1.7191381394264635. The Silhouette Coefficient is applicable only when the number of labels meets the criterion $2 \leq n_labels < n_samples - 1$ [15].

The Davies-Bouldin score is defined as the average similarity metric between each cluster and its most similar cluster, where similarity is the ratio of intra-cluster distances to inter-cluster distances. Thus, clusters that are more remote and less dispersed will receive a higher score [16]. The identified k-means clusters are shown in Figure 4.

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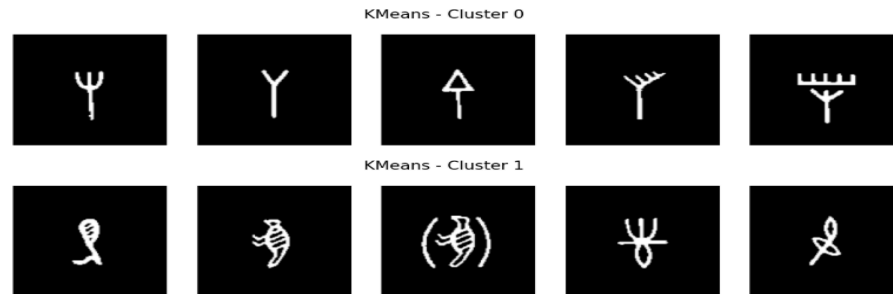


Figure 4: Visualization of K-Means algorithm-generated symbol clusters. Based on characteristics collected by ResNet50 and reduced by PCA, each cluster contains visually comparable Indus script symbols. The Silhouette Score and Davies-Bouldin Index helped one to decide that 3 is the ideal number of clusters.

5 Results and Discussion

We have used different evaluation metrics for the calculation of results. The clustering analysis indicated that K-Means was the best algorithm. It had the highest silhouette score and the lowest Davies-Bouldin index, which means that it pulled the clusters far apart and made them fit together better. The observations demonstrate that unsupervised learning can effectively study and categorize the unique patterns within the Indus script images.

The extra data added to the dataset resolved the problem of We addressed the class imbalance by creating a few more samples in the underrepresented clusters. The additional information made it possible to train the discrimination models fairly well. We have used a 70:20:10 split where 70% is training, 20% is validation, and 10% is for testing, respectively.

We have used different machine learning models for the clustering purpose of the unlabeled dataset. We have shown the score and index in Table 2. The best clustering algorithm was K-means, and the optimal number of clusters was 3.

Table 2 : Comparison of DB Index and Silhouette Score

Clustering Algorithm	Silhouette Score	Davies-Bouldin Index
KMeans	0.1273	2.2105
Agglomerative	0.1075	2.4775

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DBSCAN	Not Evaluated due to insufficient clusters	Not Evaluated due to insufficient clusters
Spectral	0.1213	2.3030
BIRCH	0.1075	2.4775

Figure 5 shows the clustering performance comparison in which we have presented the two metrics: a. Silhouette Score and b. DB Index.

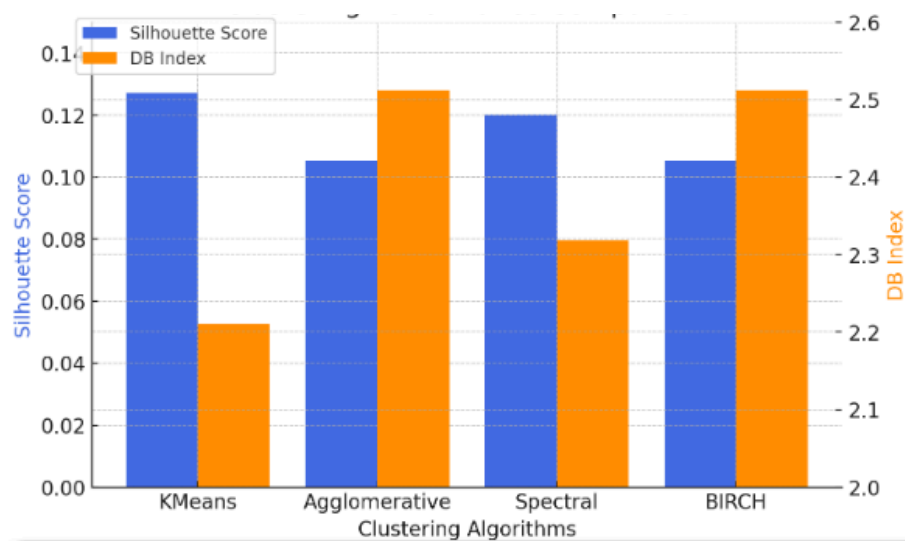


Figure 5: Based on two assessment criteria—Silhouette Score (higher is better) and Davies-Bouldin Index (lower is better)—performance comparison of many clustering techniques—KMeans, Agglomerative, Spectral, and BIRCH—based With its most favorable result, KMeans indicated well-separated and compact clusters.

After getting the labels using the K-means algorithm, which produced the best score and index, we used five different CNN models for the classification purpose. We employed 20 epochs in our work to ensure efficient training across various CNNs. We have used the k-5 fold method to validate the results obtained. Table 3 displays the performance of the deep CNN models utilized, and Figure 6 shows the graph of accuracy and loss while the deep CNN models are being validated.

Table 3 : Comparison of Accuracy and Loss

Model	Validation Accuracy	Validation Loss
VGG16	0.84	0.45

ResNet50	0.76	0.63
EfficientNet	0.75	0.66
DenseNet	0.76	0.62
MobileNet	0.82	0.51

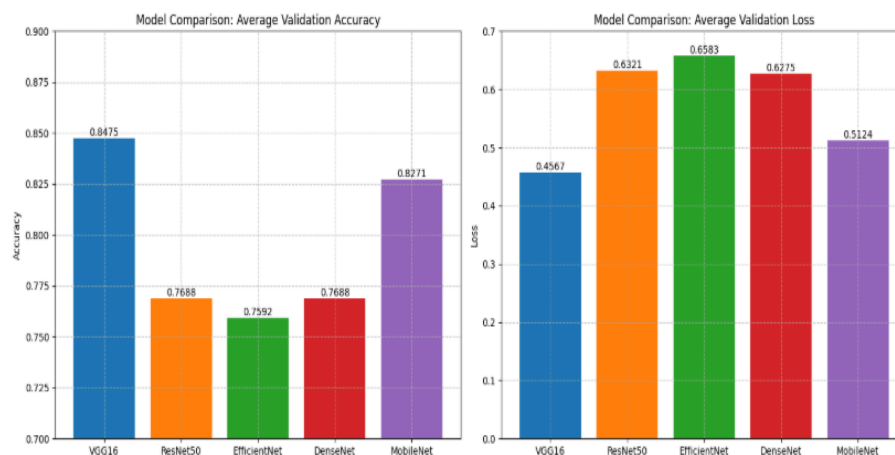


Figure 6: Comparison of Validation Accuracy and Loss of Models

As in Figure 6, we can see that VGG16 performs better than the other 4 algorithms in terms of average validation accuracy and loss.

Conclusion and Future Work

This paper demonstrates how deep learning with computer vision can be used to read and preserve ancient certifiable documents/writings, artifacts, and inscriptions. In the RAVIKANT and advancing techniques, we employed state-of-the-art neural networks to improve the accuracy and efficiency of recognizing symbols from the Indus Valley civilization, consequently moving us toward a better understanding of one of the oldest civilizations in the world.

Furthermore, this research provides a sound basis for automated recognition of past visual information, thereby paving the way for future progress in computational archaeology. The reconstruction of missing information, supporting cultural preservation, and closure between past cultures and present-day technology can all be advantageous for historians, linguists, and experts in digital heritage from different walks of life. Future research will focus on simplifying models to understand, offering more datasets, and adding multimodal AI techniques to gain a greater understanding of Indus script patterns. The R.A.V.I.K.A.N.T. technique represents a critical shift

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With the application of artificial intelligence, we are attempting to unlock the mysteries of the Indus Valley Civilization.

Acknowledgment

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